

SIA 101 – Final Project

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API Documentation

Base URL

<https://sia101-final-project.onrender.com>

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1. Authentication

1.1 JWT Authentication (User Routes)

Used for endpoints that require user identity: - GET /users/me - POST /users/me/lessons/complete - POST /users/me/quizzes/submit

Header Format:

Authorization: Bearer <jwt_token>

Token Expiration: 7 days

Token Generation: Via POST /auth/login

1.2 API Key Authentication (Service-to-Service)

Used for webhook endpoints and admin access: - POST /webhook/lesson-complete - POST /webhook/quiz-submit - GET /users/:userId

Header Format:

x-api-key: main-system-integration-key

Purpose: Secures communication between main application and progression service

2. Public Endpoints

2.1 Health Check

Endpoint: GET /health

Authentication: None

Description: Returns service status and storage information.

Request:

GET /health HTTP/1.1
Host: sia101-final-project.onrender.com

Success Response (200 OK):

```
{  
  "status": "ok",  
  "service": "progression-service",  
  "timestamp": "2026-05-21T17:00:00.000Z",  
  "storage": "in-memory"  
}
```

2.2 Service Info

Endpoint: GET /

Authentication: None

Description: Basic service information and version.

Request:

GET / HTTP/1.1
Host: sia101-final-project.onrender.com

Success Response (200 OK):

```
{
  "success": true,
  "message": "Progression Service Running",
  "version": "1.0.0",
  "storage": "in-memory (data resets on restart)"
}
```

2.3 Leaderboard

Endpoint: GET /leaderboard

Authentication: None

Description: Returns ranked users sorted by level and XP.

Query Parameters:

Parameter	Type	Required	Default	Description
limit	integer	No	10	Maximum number of users to return

Request:

GET /leaderboard?limit=5 HTTP/1.1

Host: sia101-final-project.onrender.com

Success Response (200 OK):

```
{
  "leaderboard": [
    {
      "position": 1,
      "userId": "123",
      "username": "John",
      "level": 5,
      "xp": 300,
      "totalLessons": 10,
      "totalQuizzes": 5,
      "avgQuizScore": "85.0"
    },
    {
      "position": 2,
      "userId": "456",
      "username": "Jane",
      "level": 3,
      "xp": 120,
      "totalLessons": 6,
      "totalQuizzes": 3,
      "avgQuizScore": "78.5"
    }
  ]
}
```

```
} ]
```

3. User Endpoints (JWT Protected)

3.1 Login

Endpoint: POST /auth/login

Authentication: None (public endpoint)

Description: Authenticates user and returns JWT token. Creates user if not exists.

Request Headers:

Content-Type: application/json

Request Body:

Field	Type	Required	Description
userId	string	Yes	Unique user identifier
username	string	Yes	Display name

Request Example:

```
POST /auth/login HTTP/1.1
Host: sia101-final-project.onrender.com
Content-Type: application/json
```

```
{
  "userId": "123",
  "username": "John"
}
```

Success Response (200 OK):

```
{
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "user": {
    "userId": "123",
    "username": "John",
    "level": 1,
    "xp": 0,
    "totalLessons": 0,
    "totalQuizzes": 0,
    "quizScoreSum": 0,
    "avgQuizScore": 0,
    "createdAt": "2026-05-21T17:00:00.000Z"
  }
}
```

```
}  
}
```

Error Response (400 Bad Request):

```
{  
  "error": "userId and username required"  
}
```

3.2 Get Current User Profile

Endpoint: GET /users/me

Authentication: JWT Bearer Token

Description: Retrieves logged-in user's profile with XP progress calculation.

Request Headers:

Authorization: Bearer <jwt_token>

Request Example:

```
GET /users/me HTTP/1.1  
Host: sia101-final-project.onrender.com  
Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
```

Success Response (200 OK):

```
{  
  "userId": "123",  
  "username": "John",  
  "level": 2,  
  "xp": 40,  
  "totalLessons": 1,  
  "totalQuizzes": 1,  
  "quizScoreSum": 80,  
  "avgQuizScore": 80,  
  "createdAt": "2026-05-21T17:00:00.000Z",  
  "xpNeeded": 130,  
  "progressPercent": 30  
}
```

Error Response (403 Forbidden):

```
{  
  "error": "No token provided"  
}
```

Error Response (401 Unauthorized):

```
{
  "error": "Invalid token"
}
```

Error Response (404 Not Found):

```
{
  "error": "User not found"
}
```

3.3 Complete Lesson (User Action)

Endpoint: POST /users/me/lessons/complete

Authentication: JWT Bearer Token

Description: Marks lesson as completed for current user and awards XP.

Request Headers:

Authorization: Bearer <jwt_token>

Request Example:

```
POST /users/me/lessons/complete HTTP/1.1
Host: sia101-final-project.onrender.com
Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
```

Success Response (200 OK):

```
{
  "message": "Lesson completed",
  "xpGained": 50,
  "totalLessons": 1,
  "level": 1,
  "xp": 50,
  "xpNeeded": 100,
  "levelsGained": 0
}
```

3.4 Submit Quiz (User Action)

Endpoint: POST /users/me/quizzes/submit

Authentication: JWT Bearer Token

Description: Submits quiz score for current user and calculates XP based on performance.

Request Headers:

Authorization: Bearer <jwt_token>
Content-Type: application/json

Request Body:

Field	Type	Required	Description
score	number	Yes	Points earned
maxScore	number	Yes	Maximum possible points

XP Calculation Formula: - Base XP: 30 - Score Bonus: $\text{floor}((\text{score}/\text{maxScore}) * 70)$ - Total XP: $30 + \text{scoreBonus}$

Request Example:

```
POST /users/me/quizzes/submit HTTP/1.1
Host: sia101-final-project.onrender.com
Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
Content-Type: application/json
```

```
{
  "score": 8,
  "maxScore": 10
}
```

Success Response (200 OK):

```
{
  "message": "Quiz: 80%",
  "xpGained": 86,
  "scoreBonus": 56,
  "totalQuizzes": 1,
  "avgQuizScore": "80.0",
  "level": 2,
  "xp": 36,
  "xpNeeded": 130,
  "levelsGained": 1
}
```

Error Response (400 Bad Request):

```
{
  "error": "score and maxScore required"
}
```

4. Webhook Endpoints (API Key Protected)

4.1 Lesson Completion Webhook

Endpoint: POST /webhook/lesson-complete

Authentication: API Key (x-api-key header)

Description: External system (main application) triggers lesson completion and awards XP. Creates user if not exists.

Request Headers:

x-api-key: main-system-integration-key
Content-Type: application/json

Request Body:

Field	Type	Required	Description
userId	string	Yes	Unique user identifier
username	string	Yes*	Display name (*required if new user)
lessonId	string	Yes	Lesson identifier

Request Example:

POST /webhook/lesson-complete HTTP/1.1
Host: sia101-final-project.onrender.com
x-api-key: main-system-integration-key
Content-Type: application/json

```
{
  "userId": "123",
  "username": "John",
  "lessonId": "L1"
}
```

Success Response (200 OK):

```
{
  "success": true,
  "message": "Lesson progress recorded",
  "userId": "123",
  "lessonId": "L1",
  "xpGained": 50,
  "level": 1,
  "xp": 50,
  "xpNeeded": 100,
  "levelsGained": 0
}
```

Error Response (400 Bad Request) - Missing Fields:

```
{  
  "error": "userId and lessonId required"  
}
```

Error Response (400 Bad Request) - New User Missing Username:

```
{  
  "error": "username required for new user"  
}
```

Error Response (401 Unauthorized):

```
{  
  "error": "Invalid API key"  
}
```

4.2 Quiz Submission Webhook

Endpoint: POST /webhook/quiz-submit

Authentication: API Key (x-api-key header)

Description: External system submits quiz score and awards XP based on performance. Creates user if not exists.

Request Headers:

x-api-key: main-system-integration-key
Content-Type: application/json

Request Body:

Field	Type	Required	Description
userId	string	Yes	Unique user identifier
username	string	Yes*	Display name (*required if new user)
quizId	string	Yes	Quiz identifier
score	number	Yes	Points earned
maxScore	number	Yes	Maximum possible points

Request Example:

```
POST /webhook/quiz-submit HTTP/1.1  
Host: sia101-final-project.onrender.com  
x-api-key: main-system-integration-key  
Content-Type: application/json
```

```
{
  "userId": "123",
  "username": "John",
  "quizId": "Q1",
  "score": 8,
  "maxScore": 10
}
```

Success Response (200 OK):

```
{
  "success": true,
  "message": "Quiz progress recorded",
  "userId": "123",
  "quizId": "Q1",
  "score": "80.0%",
  "xpGained": 86,
  "scoreBonus": 56,
  "level": 2,
  "xp": 36,
  "xpNeeded": 130,
  "levelsGained": 1
}
```

Error Response (400 Bad Request):

```
{
  "error": "userId, quizId, score, maxScore required"
}
```

5. Admin Endpoints (API Key Protected)

5.1 Get Any User by ID

Endpoint: GET /users/:userId

Authentication: API Key (x-api-key header)

Description: Retrieves any user's profile by their userId. Used for cross-service user lookups.

Request Headers:

x-api-key: main-system-integration-key

URL Parameters:

Parameter	Type	Required	Description
userId	string	Yes	User identifier in URL path

Request Example:

```
GET /users/123 HTTP/1.1
Host: sia101-final-project.onrender.com
x-api-key: main-system-integration-key
```

Success Response (200 OK):

```
{
  "userId": "123",
  "username": "John",
  "level": 2,
  "xp": 36,
  "totalLessons": 1,
  "totalQuizzes": 1,
  "quizScoreSum": 80,
  "avgQuizScore": 80,
  "createdAt": "2026-05-21T17:00:00.000Z",
  "xpNeeded": 130,
  "progressPercent": 27
}
```

Error Response (404 Not Found):

```
{
  "error": "User not found"
}
```

6. Error Responses

6.1 HTTP Status Codes

Status Code	Meaning	When It Occurs
200	OK	Request successful
400	Bad Request	Missing or invalid request body fields
401	Unauthorized	Invalid API key or JWT token
403	Forbidden	No authentication token provided
404	Not Found	User or resource does not exist
500	Internal Server Error	Unexpected server error

6.2 Common Error Response Format

```
{
  "error": "Descriptive error message"
}
```

6.3 Specific Error Cases

Endpoint	Error Condition	Response
All protected	Missing API Key	401 { "error": "Invalid API key" }
All JWT routes	Missing token	403 { "error": "No token provided" }
All JWT routes	Expired/invalid token	401 { "error": "Invalid token" }
/auth/login	Missing userId/username	400 { "error": "userId and username required" }
/users/me/quizzes/ submit	Missing score/maxScore	400 { "error": "score and maxScore required" }
/webhook/lesson- complete	Missing userId/lessonId	400 { "error": "userId and lessonId required" }
/webhook/quiz- submit	Missing required fields	400 { "error": "userId, quizId, score, maxScore required" }
/webhook/*	New user, no username	400 { "error": "username required for new user" }

Endpoint	Error Condition	Response
/users/:userId	User not found	404 {"error": "User not found"}
/users/me	User not found	404 {"error": "User not found"}

7. Data Models

7.1 User Object

```
{
  "userId": "string (unique identifier)",
  "username": "string (display name)",
  "level": "integer (starts at 1)",
  "xp": "integer (current XP in level)",
  "totalLessons": "integer (count of completed lessons)",
  "totalQuizzes": "integer (count of submitted quizzes)",
  "quizScoreSum": "number (sum of all quiz percentages)",
  "avgQuizScore": "number (average quiz percentage)",
  "createdAt": "Date (ISO string)"
}
```

7.2 XP System

Level Progression Formula:

XP Needed for Level = $\text{floor}(100 * 1.3^{(\text{level} - 1)})$

Level XP Requirements:

Level	XP Needed	Total XP (Cumulative)
1	100	0
2	130	100
3	169	230
4	220	399
5	286	619

XP Awards:

Action	Base XP	Bonus	Max XP
Lesson Complete	50	0	50
Quiz Submit	30	up to 70	100

Quiz Score Bonus Calculation:

```
scoreBonus = floor((score / maxScore) * 70)
totalXP = 30 + scoreBonus
```

7.3 Leaderboard Entry

```
{
  "position": "integer (rank)",
  "userId": "string",
  "username": "string",
  "level": "integer",
  "xp": "integer",
  "totalLessons": "integer",
  "totalQuizzes": "integer",
  "avgQuizScore": "string (formatted to 1 decimal)"
}
```

8. Sample Requests & Responses

8.1 Complete User Flow

Step 1: Login

POST /auth/login
Content-Type: application/json

```
{
  "userId": "student_001",
  "username": "Alice"
}
```

Response:

```
{
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VySWQiOiJzdHVkZW50XzAwMSIsInVzZXJuYW1lIjoiaWoiQWxpY2UiLCJpYXQiOiJlE3MTYzMTQwMDB9...",
  "user": {
    "userId": "student_001",
    "username": "Alice",
    "level": 1,
    "xp": 0,
  }
}
```

```
        "totalLessons": 0,  
        "totalQuizzes": 0,  
        "quizScoreSum": 0,  
        "avgQuizScore": 0,  
        "createdAt": "2026-05-21T17:00:00.000Z"  
    }  
}
```

Step 2: Complete Lesson via Webhook (Main App Calling)

POST /webhook/lesson-complete
x-api-key: main-system-integration-key
Content-Type: application/json

```
{  
  "userId": "student_001",  
  "username": "Alice",  
  "lessonId": "lesson_01"  
}
```

Response:

```
{  
  "success": true,  
  "message": "Lesson progress recorded",  
  "userId": "student_001",  
  "lessonId": "lesson_01",  
  "xpGained": 50,  
  "level": 1,  
  "xp": 50,  
  "xpNeeded": 100,  
  "levelsGained": 0  
}
```

Step 3: Submit Quiz via Webhook (Main App Calling)

POST /webhook/quiz-submit
x-api-key: main-system-integration-key
Content-Type: application/json

```
{  
  "userId": "student_001",  
  "username": "Alice",  
  "quizId": "quiz_01",  
  "score": 9,  
  "maxScore": 10  
}
```

Response:

```
{  
  "success": true,
```

```
    "message": "Quiz progress recorded",
    "userId": "student_001",
    "quizId": "quiz_01",
    "score": "90.0%",
    "xpGained": 93,
    "scoreBonus": 63,
    "level": 2,
    "xp": 43,
    "xpNeeded": 130,
    "levelsGained": 1
  }
}
```

Step 4: Check User Profile (JWT)

GET /users/me

Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...

Response:

```
{
  "userId": "student_001",
  "username": "Alice",
  "level": 2,
  "xp": 43,
  "totalLessons": 1,
  "totalQuizzes": 1,
  "quizScoreSum": 90,
  "avgQuizScore": 90,
  "createdAt": "2026-05-21T17:00:00.000Z",
  "xpNeeded": 130,
  "progressPercent": 33
}
```

Step 5: View Leaderboard

GET /leaderboard

Response:

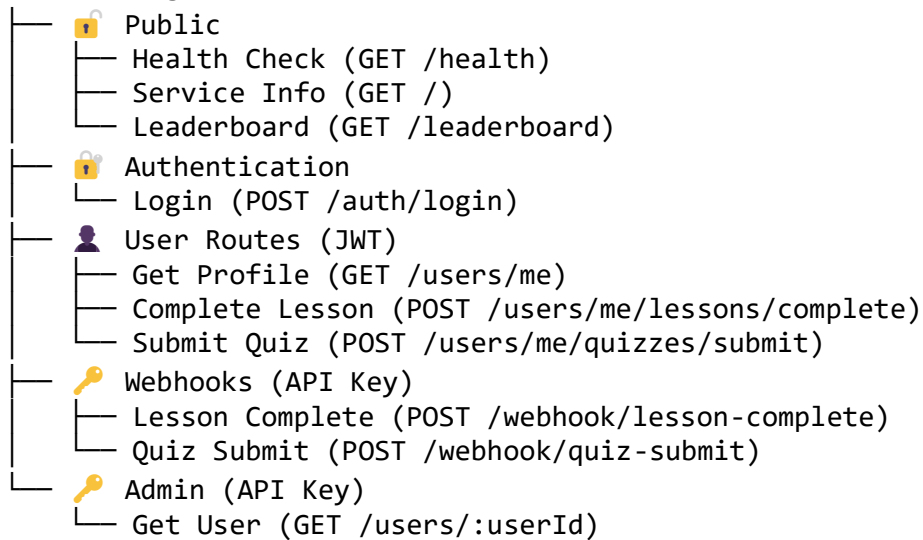
```
{
  "leaderboard": [
    {
      "position": 1,
      "userId": "student_001",
      "username": "Alice",
      "level": 2,
      "xp": 43,
      "totalLessons": 1,
      "totalQuizzes": 1,
      "avgQuizScore": "90.0"
    }
  ]
}
```

```
} ]
```

9. Postman Collection Guide

9.1 Collection Structure

SIA101 Progression Service API



9.2 Environment Variables

Variable	Value	Description
base_url	https://sia101-final-project.onrender.com	Service URL
api_key	main-system-integration-key	Service auth
jwt_token	(dynamic)	User auth token

9.3 Testing Script (for Login request)

Add this to the **Tests** tab in Postman for /auth/login:

```
var jsonData = pm.response.json();
pm.environment.set("jwt_token", jsonData.token);
```

This automatically saves the token for subsequent requests.

10. Integration Notes for Main Application

10.1 Calling Webhooks from Main App

JavaScript (Fetch):

```
const PROGRESSION_API = 'https://sia101-final-project.onrender.com';
const API_KEY = 'main-system-integration-key';

async function recordLesson(userId, username, lessonId) {
  const response = await fetch(`${PROGRESSION_API}/webhook/lesson-complete`, {
    method: 'POST',
    headers: {
      'Content-Type': 'application/json',
      'x-api-key': API_KEY
    },
    body: JSON.stringify({ userId, username, lessonId })
  });
  return response.json();
}

async function recordQuiz(userId, username, quizId, score, maxScore) {
  const response = await fetch(`${PROGRESSION_API}/webhook/quiz-submit`, {
    method: 'POST',
    headers: {
      'Content-Type': 'application/json',
      'x-api-key': API_KEY
    },
    body: JSON.stringify({ userId, username, quizId, score, maxScore })
  });
  return response.json();
}
```

10.2 Displaying Progress

After webhook calls, use the response to update UI:

```
const result = await recordLesson(userId, username, lessonId);
// result: { success, xpGained, level, xp, xpNeeded, progressPercent }

// Update UI
showNotification(`+${result.xpGained} XP!`);
updateLevelDisplay(result.level);
updateProgressBar(result.progressPercent);
```

Appendix A: Endpoint Summary

Method	Endpoint	Auth	Purpose
GET	/health	None	Service status
GET	/	None	Service info
POST	/auth/login	None	User login & token
GET	/users/me	JWT	Current user profile
POST	/users/me/lessons/complete	JWT	Mark lesson done
POST	/users/me/quizzes/submit	JWT	Submit quiz score
POST	/webhook/lesson-complete	API Key	External lesson trigger
POST	/webhook/quiz-submit	API Key	External quiz trigger
GET	/users/:userId	API Key	Lookup any user
GET	/leaderboard	None	Ranked users

Appendix B: Postman Collection

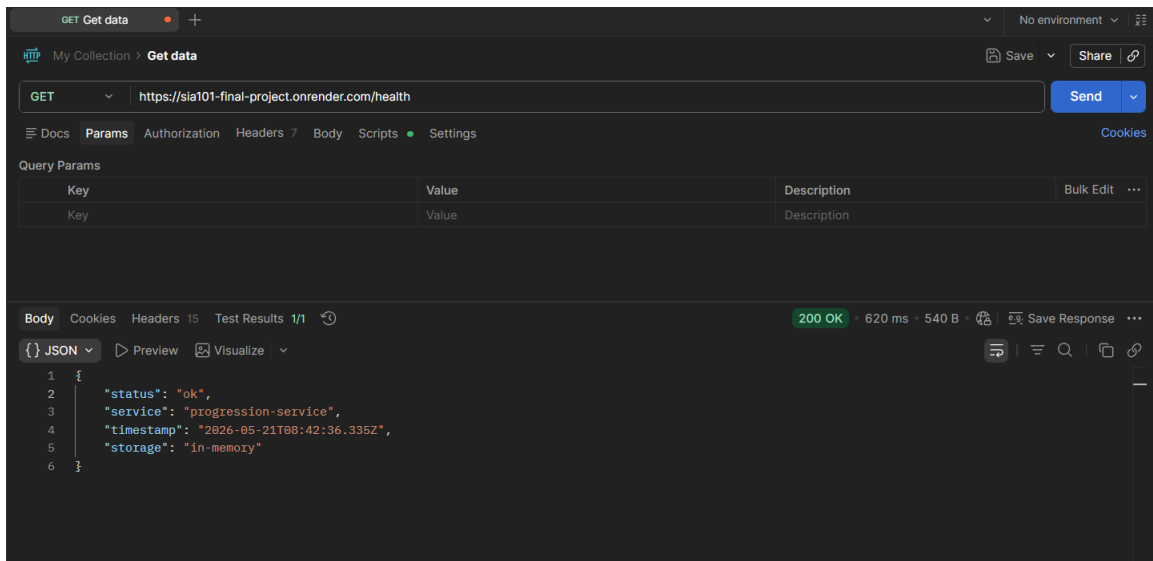
Collection Name: SIA101 Progression Service API

Export File:



SIA101_Progression_Service_Postman_Collection.json

/auth/login



/users/me

GET Get data

https://sia101-final-project.onrender.com/users/me

Send

Docs Params Authorization Headers 10 Body Scripts Settings Cookies

Headers 9 hidden

Key	Value	Description	Bulk Edit	Presets	...
Authorization	Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VyS...				
Key	Value	Description			

Body Cookies Headers 15 Test Results 1/1

200 OK · 771 ms · 602 B · Save Response

```
{
  "userId": "123",
  "username": "John",
  "level": 1,
  "xp": 0,
  "totalLessons": 0,
  "totalQuizzes": 0,
  "quizScoreSum": 0,
  "avgQuizScore": 0,
  "createdAt": "2026-05-21T08:44:55.467Z",
  "xpNeeded": 100,
  "progressPercent": 0
}
```

/users/me/lessons/complete

POST Get data

https://sia101-final-project.onrender.com/users/me/lessons/complete

Send

Docs Params Authorization Headers 10 Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON

Schema Beautify

```
{
  "score": 8,
  "maxScore": 10
}
```

Body Cookies Headers 15 Test Results 1/1

200 OK · 515 ms · 546 B · Save Response

```
{
  "message": "Lesson completed",
  "xpGained": 50,
  "totalLessons": 1,
  "level": 2,
  "xp": 36,
  "xpNeeded": 130,
  "levelsGained": 1
}
```

/users/me/quizzes/submit

The screenshot shows a REST client interface with a POST request to `https://sia101-final-project.onrender.com/users/me/quizzes/submit`. The request body is a JSON object with the following structure:

```
1 {
2   "score": 8,
3   "maxScore": 10
4 }
5
```

The response is a 200 OK status with a response time of 712 ms and a body size of 578 B. The response body is a JSON object with the following structure:

```
1 {
2   "message": "Quiz: 88%",
3   "xpGained": 86,
4   "scoreBonus": 56,
5   "totalQuizzes": 2,
6   "avgQuizScore": "88.0",
7   "level": 2,
8   "xp": 122,
9   "xpNeeded": 130,
10  "levelsGained": 0
11 }
```

/webhook/lesson-complete

The screenshot shows a REST client interface with a POST request to `https://sia101-final-project.onrender.com/webhook/lesson-complete`. The request body is a JSON object with the following structure:

```
1 {
2   "userId": "123",
3   "username": "John",
4   "lessonId": "L1"
5 }
6
```

The response is a 200 OK status with a response time of 955 ms and a body size of 574 B. The response body is a JSON object with the following structure:

```
1 {
2   "success": true,
3   "message": "Lesson progress recorded",
4   "userId": "123",
5   "lessonId": "L1",
6   "xpGained": 50,
7   "level": 3,
8   "xp": 42,
9   "xpNeeded": 169,
10  "levelsGained": 1
11 }
```

/webhook/lesson-complete

The screenshot shows a REST client interface with a POST request to `https://sia101-final-project.onrender.com/webhook/lesson-complete`. The request headers include `Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VyS...`, `x-api-key: main-system-integration-key`, and `Content-Type: application/json`. The response is a 200 OK status with a body containing the following JSON:

```
{  "success": true,  "message": "Lesson progress recorded",  "userId": "123",  "lessonId": "L1",  "xpGained": 50,  "level": 3,  "xp": 42,  "xpNeeded": 169,  "levelsGained": 1}
```

/webhook/quiz-submit

The screenshot shows a REST client interface with a POST request to `https://sia101-final-project.onrender.com/webhook/quiz-submit`. The request body is raw JSON with the following data:

```
{  "userId": "123",  "username": "John",  "quizId": "Q1",  "score": 8,  "maxScore": 10}
```

The response is a 200 OK status with a body containing the following JSON:

```
{  "success": true,  "message": "Quiz progress recorded",  "userId": "123",  "quizId": "Q1",  "score": "80.0%",  "xpGained": 86,  "scoreBonus": 56,  "level": 3,  "xp": 128,  "xpNeeded": 169,  "levelsGained": 0}
```

/users/123

GET Get data

My Collection > Get data

GET https://sia101-final-project.onrender.com/users/123

Send

Docs Params Authorization Headers 12 Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON

Schema Beautify

```
1 {
2   "userId": "123",
3   "username": "John",
4   "quizId": "Q1",
5   "score": 8,
6   "maxScore": 10
7 }
```

Body Cookies Headers 15 Test Results 1/1

200 OK · 513 ms · 613 B · Save Response

JSON Preview Visualize

```
1 {
2   "userId": "123",
3   "username": "John",
4   "level": 3,
5   "xp": 120,
6   "totalLessons": 2,
7   "totalQuizzes": 3,
8   "quizScoreSum": 240,
9   "avgQuizScore": 80,
10  "createdAt": "2026-05-21T08:44:55.467Z",
11  "xpNeeded": 169,
12  "progressPercent": 75
13 }
```

/leaderboard

GET Get data

My Collection > Get data

GET https://sia101-final-project.onrender.com/leaderboard

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7 }
```

Body Cookies Headers 15 Test Results 1/1

200 OK · 805 ms · 575 B · Save Response

JSON Preview Visualize

```
1 {
2   "leaderboard": [
3     {
4       "position": 1,
5       "userId": "123",
6       "username": "John",
7       "level": 3,
8       "xp": 120,
9       "totalLessons": 2,
10      "totalQuizzes": 3,
11      "avgQuizScore": "80.0"
12    }
13  ]
14 }
```